

A.5.2 DMS indel benchmarks

Zero-shot Table A14 shows the results for our zero-shot DMS indels benchmark, and Table A15 shows Spearman’s rank correlations for each indel DMS dataset and model. Figure A2 compares each model’s aggregate performance between the Library and Designed DMS sets (numbers provided in Table 4). More detailed performance files are available in the repository.

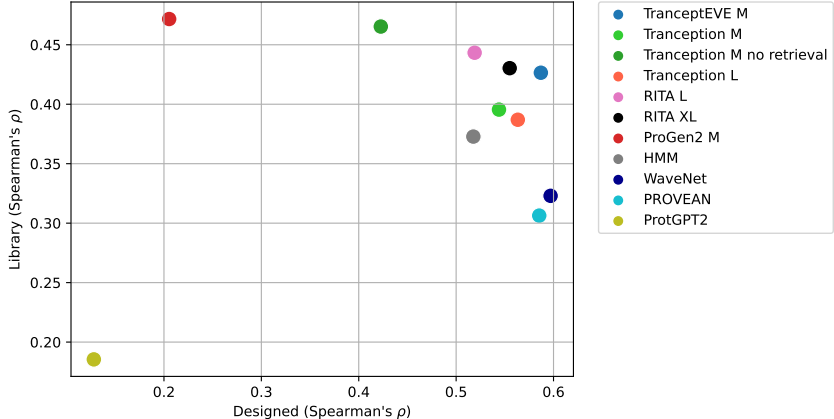


Figure A2: **Performance comparison of indel baselines on different types of assays** Spearman’s rank correlation over unbiased libraries vs model-designed sequences biased towards natural sequences.

Supervised Table A16 ranks the performance of each model on the supervised indel DMS benchmark.

A.5.3 Clinical substitution benchmarks

As discussed in § 4, since the performance of zero-shot models is on par – or higher – than their supervised counterparts we subsume the Clinical zero-shot and supervised rankings into a combined rankings, available in Table. A17. Although supervised models trained on ClinVar labels (such as ClinPred) perform well on the clinical benchmark, unsupervised models (such as TranceptEVE) provide better performance on the subset of DMS assays assessing the clinical effect of variants in humans, and competitive performance on the clinical benchmark without being subject to the same label biases (see Fig. 2).

A.5.4 Clinical indel benchmarks

Table A18 shows model performance on the ClinVar datasets, and Figure A3 shows the combined performance on the DMS and ClinVar indel benchmarks.

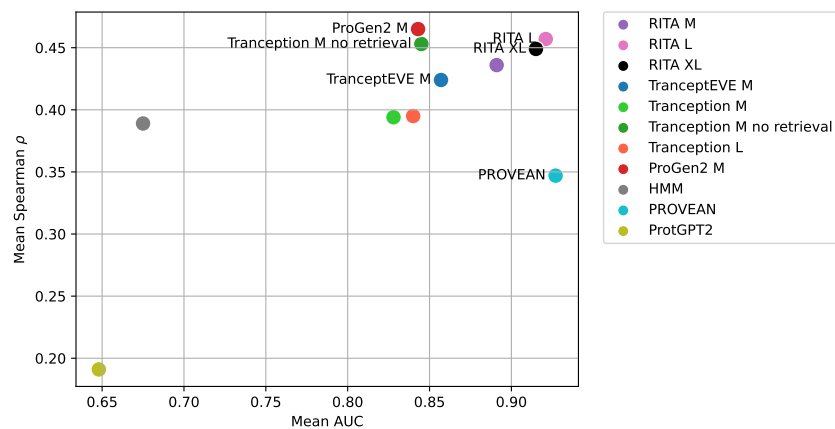


Figure A3: **Performance comparison of indel baselines on the indel benchmarks.** AUC over ClinVar with gnomAD controls (x axis) and Spearman's rank correlation over functional assay benchmarks (y axis).

Ranking	Model	Type	Spearman	Std. error
1*	TranceptEVE L	Hybrid model	0.456	0.000
1*	TranceptEVE M	Hybrid model	0.455	0.004
1*	GEMME	Alignment-based model	0.455	0.007
4	TranceptEVE S	Hybrid model	0.452	0.004
5	EVE (ensemble)	Alignment-based model	0.439	0.006
6	VESPA	Protein language model	0.436	0.006
7*	Tranception L	Hybrid model	0.434	0.004
7*	MSA Transformer (ensemble)	Hybrid model	0.434	0.009
9	EVE (single)	Alignment-based model	0.433	0.005
10	Tranception M	Hybrid model	0.427	0.005
11	ESM-IF1	Inverse folding model	0.422	0.011
12	MSA Transformer (single)	Hybrid model	0.421	0.009
13	DeepSequence (ensemble)	Alignment-based model	0.419	0.008
14	Tranception S	Hybrid model	0.418	0.006
15	ESM2 (650M)	Protein language model	0.414	0.012
16*	DeepSequence (single)	Alignment-based model	0.407	0.008
16*	ESM-1v (ensemble)	Protein language model	0.407	0.012
18	ESM2 (3B)	Protein language model	0.406	0.011
19*	MIF-ST	Inverse folding model	0.401	0.010
19*	ESM2 (15B)	Protein language model	0.401	0.010
21	EVmutation	Alignment-based model	0.395	0.006
22*	ESM-1b	Protein language model	0.394	0.010
22*	VESPA1	Protein language model	0.394	0.007
24	ProGen2 XL	Protein language model	0.391	0.008
25	ESM2 (150M)	Protein language model	0.387	0.013
26	MIF	Inverse folding model	0.382	0.011
27	ProGen2 L	Protein language model	0.380	0.008
28	ProGen2 M	Protein language model	0.379	0.008
29	ProGen2 Base	Protein language model	0.378	0.009
30*	Tranception L no retrieval	Protein language model	0.374	0.008
30*	ESM-1v (single)	Protein language model	0.374	0.013
32	WaveNet	Alignment-based model	0.373	0.012
33	RITA XL	Protein language model	0.372	0.009
34	CARP (640M)	Protein language model	0.368	0.011
35	RITA L	Protein language model	0.365	0.009
36	Site-Independent	Alignment-based model	0.359	0.010
37	RITA M	Protein language model	0.350	0.010
38	Tranception M no retrieval	Protein language model	0.348	0.009
39	Unirep evotuned	Hybrid model	0.347	0.009
40	ProGen2 S	Protein language model	0.336	0.011
41	CARP (76M)	Protein language model	0.328	0.012
42	ESM2 (35M)	Protein language model	0.321	0.015
43	RITA S	Protein language model	0.304	0.011
44	Tranception S no retrieval	Protein language model	0.303	0.012
45	CARP (38M)	Protein language model	0.279	0.014
46	ProteinMPNN	Inverse folding model	0.258	0.011
47	ESM2 (8M)	Protein language model	0.226	0.015
48	UniRep	Protein language model	0.190	0.016
49	ProtGPT2	Protein language model	0.188	0.011
50	CARP (600K)	Protein language model	0.106	0.016

Table A5: **ProteinGym - Zero-shot substitution DMS benchmark** Ranking based on corrected average of Spearman’s rank correlation between experimental assay measurement and model prediction. The standard error reported corresponds to the non-parametric bootstrap standard error of the difference between the Spearman performance of a given model and that of the best overall model (i.e., TranceptEVE), computed over 10k bootstrap samples from the set of proteins in the ProteinGym DMS substitution benchmark.

Model type	Model name	Spearman by MSA depth ($\uparrow$)			
		Low	Medium	High	All
Alignment-based	Site-Independent	0.426	0.373	0.320	0.373
	WaveNet	0.299	0.389	0.452	0.380
	EVmutation	0.403	0.423	0.410	0.412
	DeepSequence (ens.)	0.383	0.428	0.473	0.428
	EVE (ens.)	0.425	0.453	0.481	0.453
	GEMME	0.455	0.470	0.497	0.474
Protein language	UniRep	0.181	0.161	0.209	0.184
	CARP (640M)	0.314	0.375	0.428	0.372
	ESM-1b	0.350	0.398	0.482	0.410
	ESM-2 (15B)	0.357	0.414	0.473	0.415
	RITA XL	0.315	0.382	0.412	0.370
	ESM-1v (ens.)	0.326	0.418	0.502	0.415
	ProGen2 XL	0.354	0.405	0.444	0.401
	VESPA	0.427	0.455	0.484	0.455
Hybrid	UniRep evotuned	0.330	0.344	0.372	0.349
	MSA Transformer (ens.)	0.404	0.450	0.488	0.447
	Tranception L	0.432	0.438	0.473	0.448
	TranceptEVE L	0.451	0.467	0.492	0.470
Inverse Folding	ESM-IF1	0.300	0.431	0.544	0.425
	MIF-ST	0.376	0.403	0.456	0.412
	ProteinMPNN	0.173	0.280	0.434	0.296

Table A6: **ProteinGym - Zero-shot substitution DMS benchmark by MSA depth** Average Spearman’s rank correlation between model scores and experimental measurements by MSA depth on the ProteinGym substitution benchmark. Alignment depth is measured by the ratio of the effective number of sequences N_{eff} in the MSA, following Hopf et al. [2017], by the length covered L (Low: $N_{\text{eff}}/L < 1$; Medium: $1 < N_{\text{eff}}/L < 100$; High: $N_{\text{eff}}/L > 100$). The All column is the average across the 3 depths.

Model type	Model name	Spearman by Function Type ($\uparrow$)					
		Activity	Binding	Expression	Organismal Fitness	Stability	All
Alignment-based	Site-Independent	0.369	0.345	0.351	0.382	0.358	0.361
	WaveNet	0.379	0.325	0.350	0.365	0.449	0.374
	EVmutation	0.440	0.322	0.382	0.411	0.430	0.397
	DeepSequence (ens.)	0.455	0.368	0.396	0.413	0.476	0.422
	EVE (ens.)	0.464	0.394	0.406	0.447	0.491	0.440
	GEMME	0.482	0.387	0.443	0.452	0.519	0.457
Protein language	UniRep	0.182	0.203	0.230	0.141	0.210	0.193
	CARP (640M)	0.395	0.274	0.419	0.364	0.414	0.373
	ESM-1b	0.428	0.289	0.427	0.351	0.500	0.399
	ESM-2 (15B)	0.405	0.318	0.425	0.388	0.488	0.405
	RITA XL	0.366	0.303	0.416	0.381	0.398	0.373
	ESM-1v (ens.)	0.414	0.320	0.456	0.387	0.500	0.415
	ProGen2 XL	0.402	0.302	0.423	0.387	0.445	0.392
	VESPA	0.468	0.365	0.410	0.440	0.500	0.437
Hybrid	UniRep evotuned	0.355	0.304	0.366	0.346	0.366	0.347
	MSA Transformer (ens.)	0.469	0.343	0.439	0.421	0.495	0.433
	Tranception L	0.465	0.351	0.455	0.436	0.471	0.436
	TranceptEVE L	0.487	0.381	0.456	0.460	0.500	0.457
Inverse Folding	ESM-IF1	0.368	0.392	0.403	0.324	0.624	0.422
	MIF-ST	0.390	0.323	0.432	0.373	0.486	0.401
	ProteinMPNN	0.197	0.165	0.198	0.165	0.566	0.258

Table A7: **ProteinGym - Zero-shot substitution DMS benchmark by function type** Corrected average of Spearman’s rank correlation between model scores and experimental measurements on the ProteinGym substitution benchmark, separated into five functional categories (Activity, Binding, Organismal Fitness, Stability and Expression). ‘All’ is the average of all the categories.

Model type	Model name	Spearman by Taxa ($\uparrow$)				
		Human	Other Eukaryote	Prokaryote	Virus	All
Alignment-based	Site-Independent	0.379	0.385	0.316	0.383	0.366
	WaveNet	0.391	0.410	0.427	0.328	0.389
	EVmutation	0.409	0.444	0.422	0.388	0.416
	DeepSequence (ens.)	0.442	0.469	0.460	0.344	0.429
	EVE (ens.)	0.453	0.487	0.468	0.428	0.459
	GEMME	0.468	0.510	0.473	0.469	0.480
Protein language	UniRep	0.213	0.219	0.165	0.057	0.164
	CARP (640M)	0.416	0.386	0.390	0.273	0.366
	ESM-1b	0.434	0.475	0.455	0.241	0.401
	ESM-2 (15B)	0.431	0.449	0.459	0.313	0.413
	RITA XL	0.394	0.384	0.353	0.402	0.383
	ESM-1v (ens.)	0.458	0.446	0.454	0.289	0.412
	ProGen2 XL	0.384	0.442	0.439	0.391	0.414
	VESPA	0.438	0.492	0.490	0.432	0.463
Hybrid	UniRep evotuned	0.355	0.363	0.346	0.349	0.353
	MSA Transformer (ens.)	0.437	0.505	0.463	0.414	0.455
	Tranception L	0.453	0.483	0.431	0.432	0.450
	TranceptEVE L	0.471	0.498	0.473	0.453	0.474
Inverse Folding	ESM-IF1	0.415	0.497	0.507	0.374	0.448
	MIF-ST	0.404	0.415	0.463	0.396	0.420
	ProteinMPNN	0.282	0.395	0.354	0.248	0.320

Table A8: **ProteinGym - Zero-shot substitution DMS benchmark by taxa** Average Spearman’s rank correlation between model scores and experimental measurements on the ProteinGym substitution benchmark, separated by taxon. ‘All’ is the average across the taxa.